Spectral Theorem for Real Symmetric Matrices

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of spectral theorem for real symmetric matrices, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

LA-03 / Linear Algebra / Report v1

I. TARGET AND IMPORT CONTRACT

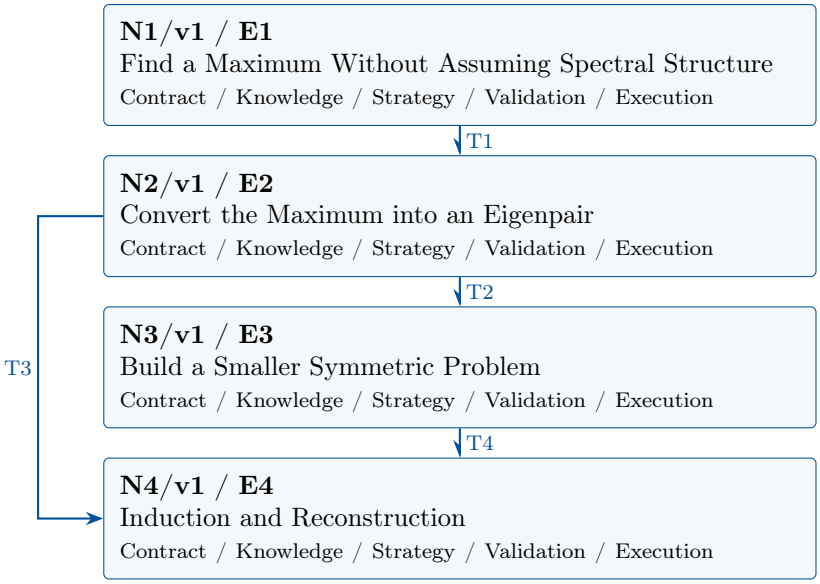
Theorem I.1. *For every real symmetric $n \times n$ matrix A, there are a real orthogonal matrix Q and a real diagonal matrix Lambda such that $A = Q\Lambda Q^T$.*

Equivalently, Euclidean space has an orthonormal basis of eigenvectors of A. The output includes real eigenvalues, orthogonal eigendirections, and a reconstruction certificate. It does not claim uniqueness of an eigenbasis within a repeated eigenspace.

A. Foundations and permitted imports

A is symmetric when $A^T = A$; Q is orthogonal when $Q^TQ = I$. Import compactness of the finite-dimensional unit sphere, the extreme value theorem, elementary differentiation, and induction on dimension. No diagonalization theorem is imported.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The Rayleigh quotient attains a real maximum at a unit vector v.

Receiving activity. N2: Convert the Maximum into an Eigenpair

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and differentiable tangent curves on the sphere.

T2: N2/E2/v1 to N3. $Av = \lambda v$ with $\lambda = v^T Av \in \mathbb{R}$.

Receiving activity. N3: Build a Smaller Symmetric Problem

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2 and symmetry of the original operator.

T3: N2/E2/v1 to N4. $Av = \lambda v$ with $\lambda = v^T Av \in \mathbb{R}$.

Receiving activity. N4: Induction and Reconstruction

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N3/E3, and induction on dimension.

T4: N3/E3/v1 to N4. $v^\perp$ is invariant under A, and the restriction there is symmetric of dimension $n-1$.

Receiving activity. N4: Induction and Reconstruction

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N3/E3, and induction on dimension.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Find a Maximum Without Assuming Spectral Structure

Object and role. Find a Maximum Without Assuming Spectral Structure. Establish the following local result: The Rayleigh quotient attains a real maximum at a unit vector v.

Inputs. A real symmetric $n \times n$ matrix, $n \geq 1$; compactness of the unit sphere.

C — Contract

Own E1/v1: The Rayleigh quotient attains a real maximum at a unit vector v.

K — Knowledge

Symmetry and Euclidean inner product

S — Strategy

Study quadratic form on unit sphere

V — Validation

Require finite dimension and real symmetry

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Restrict $x^T Ax$ to the unit sphere.

N1.2 Check continuity and apply the extreme value theorem.

N1.3 Fix a maximizing unit vector without assuming any spectral decomposition.

Export E1/v1. The Rayleigh quotient attains a real maximum at a unit vector v.

Acceptance gate. Require finite dimension and real symmetry Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Require finite dimension and real symmetry Recheck dependent nodes N2, N3, N4.

B. N2/v1: Convert the Maximum into an Eigenpair

Object and role. Convert the Maximum into an Eigenpair. Establish the following local result: $Av = \lambda v$ with $\lambda = v^T Av \in \mathbb{R}$.

Inputs. N1/E1 and differentiable tangent curves on the sphere.

C — Contract

Own E2/v1: $Av = \lambda v$ with $\lambda = v^T Av \in \mathbb{R}$.

K — Knowledge

Maximum and tangent curves

S — Strategy

Differentiate at maximizing point

V — Validation

Avoid assuming eigenvector existence

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For each $w \perp v$, differentiate along $(v + tw)/\sqrt{1 + t^2\|w\|^2}$.

N2.2 The derivative vanishes at the maximum, giving $w^T Av = 0$.

N2.3 Therefore Av lies in the line spanned by v.

Export E2/v1. $Av = \lambda v$ with $\lambda = v^T Av \in \mathbb{R}$.

Acceptance gate. Avoid assuming eigenvector existence Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Avoid assuming eigenvector existence Recheck dependent nodes N3, N4.

C. N3/v1: Build a Smaller Symmetric Problem

Object and role. Build a Smaller Symmetric Problem. Establish the following local result: $v^{\perp}$ is invariant under A, and the restriction there is symmetric of dimension $n-1$.

Inputs. N2/E2 and symmetry of the original operator.

C — Contract

Own E3/v1: $v^{\perp}$ is invariant under A, and the restriction there is symmetric of dimension $n-1$.

K — Knowledge

Symmetric inner-product identity

S — Strategy

Reduce dimension orthogonally

V — Validation

Restriction must remain invariant and symmetric

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 For $w \perp v$, compute $\langle Aw, v\rangle = \langle w, Av\rangle = 0$.

N3.2 Restrict the symmetric inner-product identity to the invariant subspace.

N3.3 Prepare the smaller-dimensional induction problem.

Export E3/v1. $v^{\perp}$ is invariant under A, and the restriction there is symmetric of dimension $n-1$.

Acceptance gate. Restriction must remain invariant and symmetric Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Restriction must remain invariant and symmetric Recheck dependent nodes N4.

D. N4/v1: Induction and Reconstruction

Object and role. Induction and Reconstruction. Establish the following local result: $A = Q\Lambda Q^T$ for an orthogonal Q and real diagonal Λ.

Inputs. N2/E2, N3/E3, and induction on dimension.

C — Contract

Own E4/v1: $A = Q\Lambda Q^T$ for an orthogonal Q and real diagonal Λ.

K — Knowledge

Lower-dimensional result

S — Strategy

Combine orthonormal bases

V — Validation

Handle repeated eigenvalues

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Apply the theorem inductively on $v^\perp$, starting with dimension one.

N4.2 Join its orthonormal eigenbasis with v.

N4.3 Check $Q^T AQ = \Lambda$ and the reconstruction $A = \sum_i \lambda_i v_i v_i^T$.

Export E4/v1. $A = Q\Lambda Q^T$ for an orthogonal Q and real diagonal Λ.

Acceptance gate. Handle repeated eigenvalues Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Handle repeated eigenvalues Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Find a Maximum Without Assuming Spectral Structure

For $n \geq 1$, the function $R(x) = x^T Ax$ is continuous on the compact unit sphere. It attains a maximum at a unit vector v. Set $\lambda = v^T Av$, which is real. This constructs a candidate eigenvector through optimization rather than assuming one exists.

B. N2 — Convert the Maximum into an Eigenpair

Let w be any vector perpendicular to v, and consider

$$x(t) = \frac{v + tw}{\sqrt{1 + t^2\|w\|^2}}.$$

This stays on the unit sphere, $x(0) = v$, and x'(0) $= w$. Since $R(x(t))$ has a maximum at zero, its derivative there vanishes. Symmetry gives

$$0 = \frac{d}{dt} x(t)^T Ax(t)\bigg|_{t=0} = w^T Av + v^T Aw = 2w^T Av.$$

Thus Av is orthogonal to every vector in v-perp, so Av is a multiple of v. Taking its inner product with v gives $Av = \lambda v$. This establishes a real eigenpair for every nonzero-dimensional symmetric operator.

C. N3 — Build a Smaller Symmetric Problem

For u in v-perp,

$$v^T Au = (Av)^T u = \lambda v^T u = 0.$$

Hence Au remains in v-perp: this subspace is invariant. The restricted operator is symmetric because $\langle Au, w\rangle = \langle u, Aw\rangle$ still holds for u,w there.

The restriction acts on an $(n-1)$-dimensional Euclidean space. The proof has now generated a smaller instance of the same theorem, with the hypotheses preserved.

D. N4 — Induction and Reconstruction

For $n = 1$ the theorem is immediate. Assume it in dimension $n - 1$. Apply it to the N3 restriction to obtain an orthonormal eigenbasis of v-perp. Adjoin v to get an orthonormal eigenbasis $v_1, \ldots, v_n$ of the original space.

Let Q have those vectors as columns and let Lambda contain their eigenvalues. Then $AQ = Q$ Lambda and $Q^T Q = I$. Thus $Q^T AQ = \Lambda$ and $A = Q\Lambda Q^T$. This closes the induction. □

As a further consequence,

$$A = \sum_{j=1}^{n} \lambda_j v_j v_j^T.$$

Different eigenvalues have perpendicular eigenvectors: if $Au = \lambda$ u and $Av = mu$ v, symmetry gives $(\lambda - \mu)u^T v = 0$. Repeated eigenvalues are handled by an orthonormal basis within their eigenspace; they do not obstruct the theorem.

□

V. WORKED ORCHESTRATION

Take

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

The quadratic form is $2x^2 + 2xy + 2y^2 + 4z^2$. Since $2xy \le x^2 + y^2$, it is at most $3(x^2 + y^2) + 4z^2 \le 4$ on the unit sphere, with equality at e_3. N2 therefore supplies eigenpair $(4, e_3)$.

N3 restricts to the xy-plane. Its matrix is [[2,1],[1,2]], with orthonormal eigenvectors

$$v_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad \lambda_1 = 3; \qquad v_2 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \lambda_2 = 1.$$

Together with $v_3 = e_3$ and lambda $_3 = 4$, take

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \Lambda = \mathrm{diag}(3, 1, 4).$$

Execution verifies $Q^T Q = I$, $Av_j = \lambda_j\ v_j$, and the reconstruction. The first two spectral contributions are

$$3v_1 v_1^T + v_2 v_2^T = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

while $4v_3 v_3^T$ supplies the last diagonal entry.

A. Boundary and hypothesis audit

Symmetry is essential: [[1,1],[0,1]] has only one independent eigenvector and cannot be diagonalized. A real rotation may have no real eigenvalues. Positive eigenvalues are not

required: symmetric matrices can have negative eigenvalues. The zero matrix is covered and permits any orthonormal basis.

Export the extremal-eigenvector lemma, invariant-complement lemma, orthogonal diagonalization, and rank-one spectral reconstruction. These are the exact imports used to develop singular value decomposition in the problem bank.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

Suppose a branch normalizes several independently chosen eigenvectors and assumes they are orthogonal. For $A = I$, e_1 and $e_1 + e_2$ are both eigenvectors, but normalization alone does not make them orthogonal.

- **Node:** N4
- **State:** STALLED
- **Failing component:** Execution
- **Failed claim:** normalized eigenvectors form orthogonal Q
- **Observed:** off-diagonal entry of Q transpose Q is nonzero
- **Repair target:** basis selection
- **Blocked export:** orthogonal diagonalization certificate

Knowledge recalls orthogonal complements and repeated eigenspaces. Validation checks whether the chosen vectors are independent. Strategy constructs an orthonormal basis in the repeated eigenspace. Execution verifies its inner products and that every new vector remains an eigenvector. Contract republishes the decomposition only after $Q^TQ = I$ holds.

B. Hypothesis boundary

Symmetry is essential: [[1,1],[0,1]] has only one independent eigenvector and cannot be diagonalized. A real rotation may have no real eigenvalues. Positive eigenvalues are not required: symmetric matrices can have negative eigenvalues. The zero matrix is covered and permits any orthonormal basis.

Export the extremal-eigenvector lemma, invariant-complement lemma, orthogonal diagonalization, and rank-one spectral reconstruction. These are the exact imports used to develop singular value decomposition in the problem bank.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. MIT linear algebra: proof of the spectral theorem.